

# Two-Phase PageRank with Travel-Time Self-Loops

Traffic Prediction on the Salt Lake City Road Network

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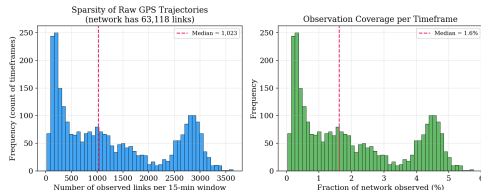
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# Outline

- 1 Problem & Approach
- 2 Building the Transition Matrix  $M_{2N}$
- 3 Building the Teleportation Vector  $E_N$
- 4 The Power Iteration
- 5 Smoothing Justification
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# The Traffic Prediction Problem

- Salt Lake City: **99,716 directed road links**
- GPS traces cover only a **tiny fraction** of the network per 15-min window
- Goal: predict the **full traffic distribution** across every road

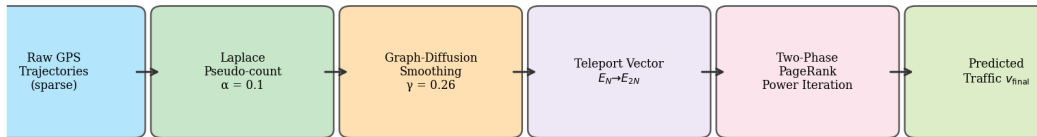


## Our Approach

Model traffic as a **two-phase random walk** on the road graph, incorporating physical travel time

# End-to-End Pipeline

Two-Phase PageRank: End-to-End Data Pipeline



# Two Components of the Model

**Power Iteration:**  $\mathbf{v}^{(k+1)} = \underbrace{d \cdot M_{2N}^T \cdot \mathbf{v}^{(k)}}_{\text{Transition Matrix (topology)}} + \underbrace{(1 - d) \cdot E_{2N}}_{\text{Teleportation Vector (ground truth)}}$

| Component                     | Built from                     | Role                           |
|-------------------------------|--------------------------------|--------------------------------|
| Transition matrix $M_{2N}$    | Road topology, lengths, speeds | Where vehicles <b>move</b>     |
| Teleportation vector $E_{2N}$ | Ground-truth GPS counts        | Where vehicles are <b>born</b> |

# The Two Phases of Driving

Every road link  $i$  has **two states**:

## Up Phase

Vehicle is **actively traversing** link  $i$

Stays on link proportional to  
its physical travel time

**Self-loop traps mass**



## Down Phase

Vehicle is **locally dispersing**

Moves to adjacent links  
in the local neighbourhood

Represents turning, parking, etc.

$\beta = 0.90$ : probability of dropping from Up to Down per step

# Up Phase: Travel-Time Self-Loops

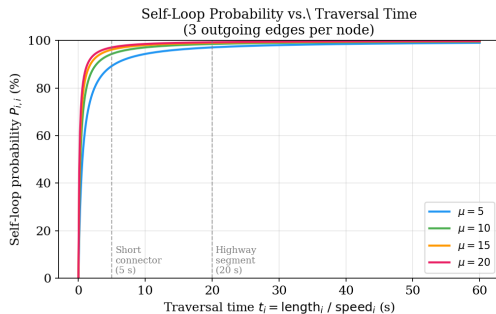
A vehicle on link  $i$  has traversal time  $t_i = \ell_i / s_i$  seconds.

We add a **self-loop** so the vehicle “dwells” on the link proportionally:

$$w_i = \mu \cdot \frac{\ell_i}{s_i}$$

- $\ell_i$  = road length,  $s_i$  = speed limit
- $\mu = 20$  = dwell-time scale

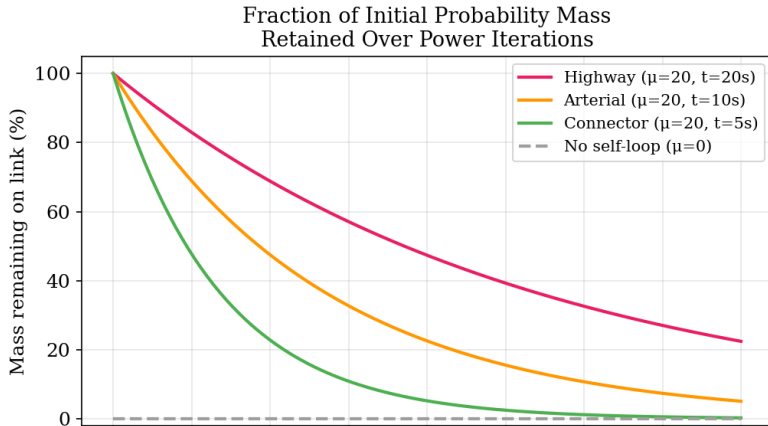
**Longer/slower roads retain more mass**



# Self-Loop Probability

After adding the self-loop, the row-stochastic transition for link  $i$  becomes:

$$P_{\text{self}}(i, i) = \frac{w_i}{w_i + \deg^+(i)}, \quad P_{\text{self}}(i, j) = \frac{A_{ij}}{w_i + \deg^+(i)}$$

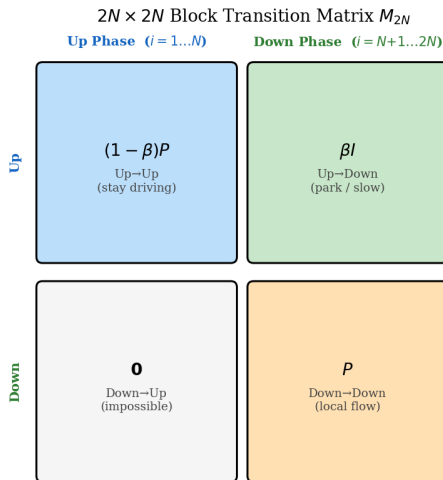




# The $2N \times 2N$ Block Matrix

$$M_{2N} = \begin{pmatrix} (1-\beta) P_{\text{self}} & \beta I \\ 0 & P_{\text{adj}} \end{pmatrix}$$

- **Top-left:** Up→Up  
(stay driving, with self-loops)
- **Top-right:** Up→Down  
(drop to local dispersal)
- **Bottom-right:** Down→Down  
(move to adjacent links)
- **Bottom-left:** 0  
(no return from Down to Up)



## Key Fact: $M_{2N}$ Uses No Ground Truth

$M_{2N}$  is built **entirely** from the **road network**: topology, lengths, and speed limits.  
**No GPS data. No traffic counts.**

The transition matrix defines the “physics”—  
the rules of how vehicles move through the network.

It is **built once** for a city and **reused** for every 15-minute prediction window.

# Four-Step Construction of $E_N$

- ① **Extract raw GPS counts**  $\mathbf{c} \in \mathbb{Z}_{\geq 0}^N$  (most  $c_i = 0$ )
- ② **Laplace pseudo-count:**  $\tilde{c}_i = c_i + 0.1$  (eliminate zeros)
- ③ **Graph-diffusion smoothing** ( $\gamma = 0.26$ ):

$$\mathbf{c}_{\text{smooth}} = 0.74 \tilde{\mathbf{c}} + 0.26 P_{\text{adj}} \tilde{\mathbf{c}}$$

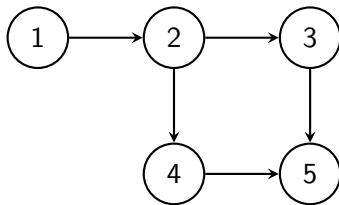
- ④  **$L^1$ -normalise:**

$$E_N[i] = \frac{c_{\text{smooth},i}}{\sum_j c_{\text{smooth},j}}, \quad \sum_i E_N[i] = 1$$

## Result

$E_N[i]$  = probability that a new vehicle appears on link  $i$

## Worked Example: 5-Link Network



|                         | Link 1       | Link 2       | Link 3       | Link 4       | Link 5       |
|-------------------------|--------------|--------------|--------------|--------------|--------------|
| Raw count $c_i$         | 0            | 5            | 0            | 3            | 0            |
| + Laplace $\tilde{c}_i$ | 0.10         | 5.10         | 0.10         | 3.10         | 0.10         |
| Neighbour avg           | 5.10         | 1.60         | 0.10         | 0.10         | —            |
| Smoothed                | 1.40         | 4.19         | 0.10         | 2.32         | 0.10         |
| $E_N[i]$                | <b>0.172</b> | <b>0.516</b> | <b>0.012</b> | <b>0.286</b> | <b>0.012</b> |

Link 1 gets **17.2%** despite zero observations  $\Rightarrow$  topology propagates evidence from its busy neighbour

## Extending to $2N$ : The $E_{2N}$ Vector

Teleportation injects vehicles only into the **Up (active)** phase:

$$E_{2N} = \begin{pmatrix} E_N \\ \mathbf{0}_N \end{pmatrix} \in \mathbb{R}^{2N}, \quad \|E_{2N}\|_1 = 1$$

### Physical meaning:

- New vehicles are “born” actively driving on a road
- They naturally transition to the Down phase over time
- The ground truth determines *where* they appear; the transition matrix determines *how* they move

# Combining Topology + Ground Truth

## The Power Iteration

$$\mathbf{v}^{(k+1)} = \underbrace{d \cdot M_{2N}^T \cdot \mathbf{v}^{(k)}}_{\text{topology-driven flow (80\%)}} + \underbrace{(1-d) \cdot E_{2N}}_{\text{ground-truth injection (20\%)}}$$

$$d = 0.80$$

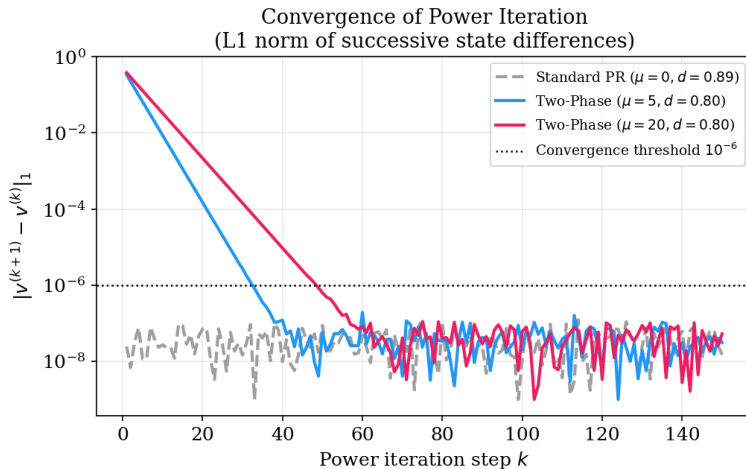
80% of mass follows the  
**transition matrix**  
(roads, travel times, phases)

$$1 - d = 0.20$$

20% of mass is re-injected  
from **ground-truth GPS**  
(observed traffic pattern)

Converges when  $\|\mathbf{v}^{(k+1)} - \mathbf{v}^{(k)}\|_1 < 10^{-6}$

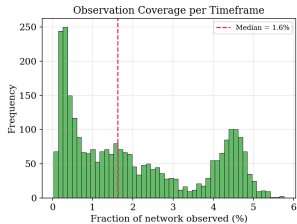
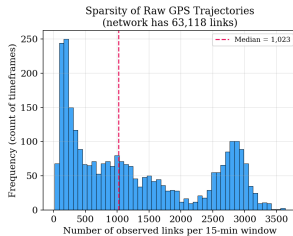
# Convergence



Typically converges in  $\sim 40$ – $60$  iterations

# Why Smooth? — The Sparsity Problem

- Most links have  $c_i = 0$
- Creates:
  - MRE singularities
  - Markov chain dead zones
  - Biased teleportation
- Without smoothing:  
**MRE = 7,275**
- With smoothing:  
**MRE = 0.19**





# Graph-Diffusion as a Low-Pass Filter

$$\mathbf{c}_{\text{smooth}} = \underbrace{(1 - \gamma)}_{\text{keep own}} \cdot \tilde{\mathbf{c}} + \underbrace{\gamma}_{\text{borrow from neighbours}} \cdot P_{\text{adj}} \tilde{\mathbf{c}}$$

## Spectral view:

Eigenvalue of the filter:

$$h(\lambda) = (1 - \gamma) + \gamma\lambda$$

$$h(1) = 1 \text{ (preserves mean)}$$

$$h(-1) = 1 - 2\gamma \text{ (attenuates noise)}$$

## Mass conservation:

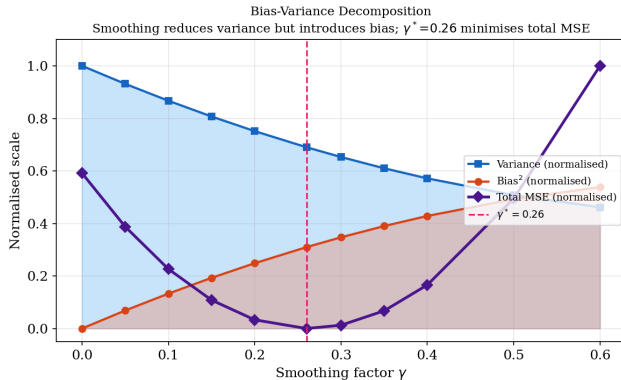
$$\sum_i c_{\text{smooth},i} = \sum_i \tilde{c}_i$$

Smoothing **redistributes**  
mass but never creates  
or destroys it

# The Bias-Variance Tradeoff

$$\text{MSE} = \underbrace{\text{Bias}^2}_{\uparrow \text{ with } \gamma} + \underbrace{\text{Variance}}_{\downarrow \text{ with } \gamma}$$

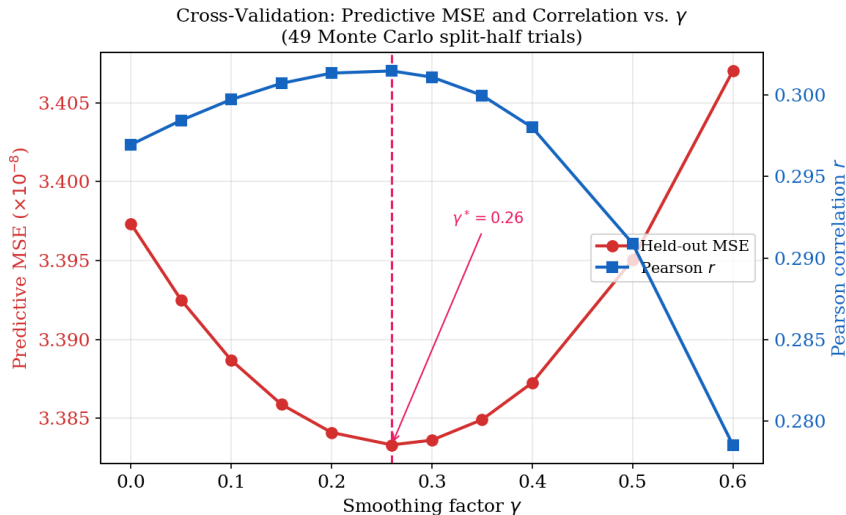
- Too little smoothing → **noisy**
- Too much smoothing → **blurred**
- U-shaped MSE curve
- Minimum **inside** the range



# Cross-Validation Protocol (49 Trials)

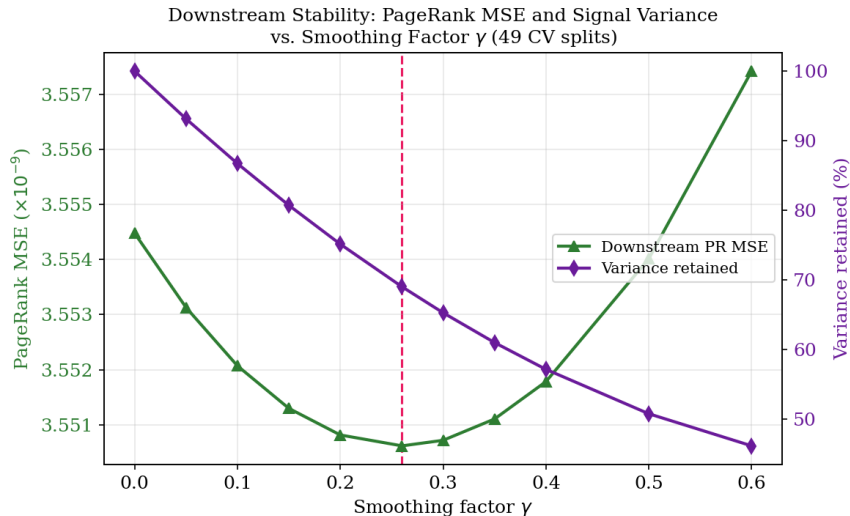
- 1 Select 49 timeframe slots (fixed seed = 42)
- 2 **Split** GPS data: Set A (train) / Set B (test)
- 3 Smooth Set A with candidate  $\gamma$
- 4 **Never** smooth Set B (oracle ground truth)
- 5 Measure:
  - **Predictive MSE**: smoothed A vs. raw B
  - **PageRank MSE**: PageRank(A) vs. PageRank(B)
  - **Pearson  $r$** : correlation A vs. B
- 6 Average over all 49 trials

# CV Results: MSE & Correlation



Both metrics optimised at  $\gamma^* = 0.26$

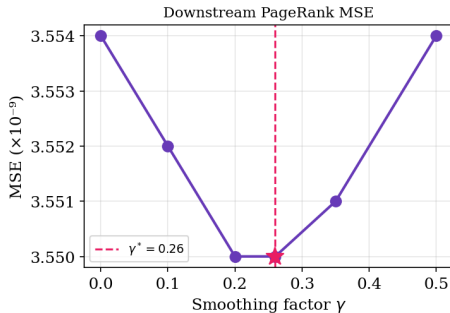
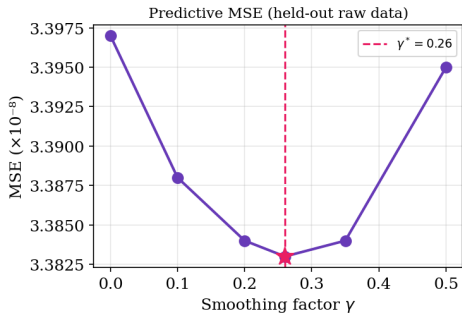
# CV Results: Downstream PageRank MSE



PageRank MSE minimum **also** at  $\gamma \approx 0.26 \Rightarrow$  input-optimal = output-optimal

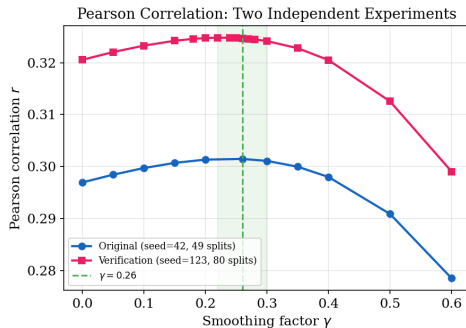
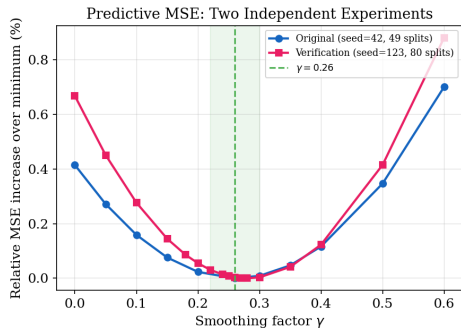
# Direct vs. Downstream MSE

Cross-Validation: Optimal Smoothing Parameter ( $\gamma^* = 0.26$ )



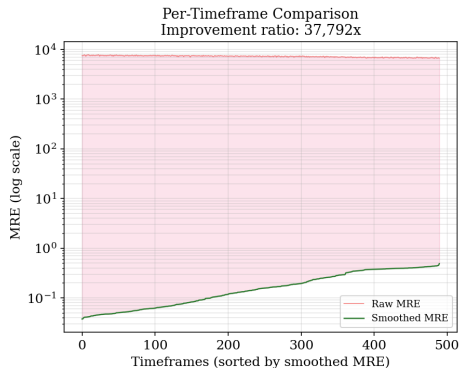
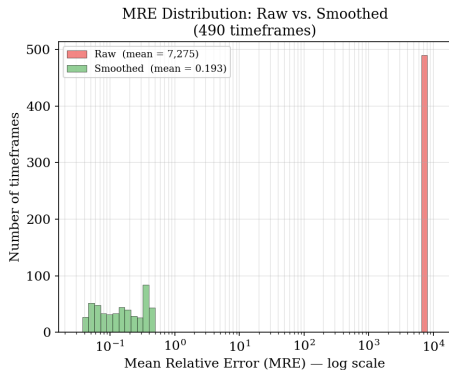
# Independent Verification (Seed = 123, 80 Splits)

Independent Verification of Optimal  $\gamma^*$   
(green band = optimal plateau  $\gamma \in [0.22, 0.30]$ )



Broad plateau at  $\gamma \in [0.22, 0.30]$ ;  $\gamma = 0.26$  is the **centre of the plateau**.  
MSE varies  $< 0.04\%$  within this band.

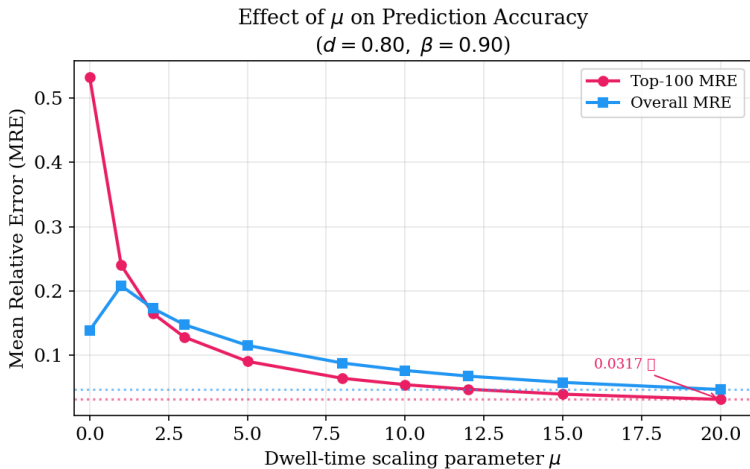
# Impact of Smoothing: 490-Timeframe Comparison



| Pipeline                     | Avg MRE | Improvement     |
|------------------------------|---------|-----------------|
| Raw ( $\gamma = 0$ )         | 7,275   | —               |
| Smoothed ( $\gamma = 0.26$ ) | 0.19    | $\times 37,800$ |

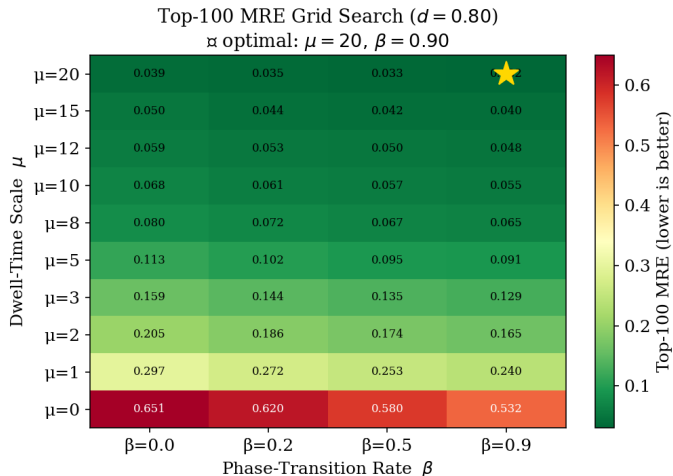


# Effect of $\mu$ (Dwell-Time Scale)



MRE drops monotonically as  $\mu$  increases. Optimal:  $\mu = 20$

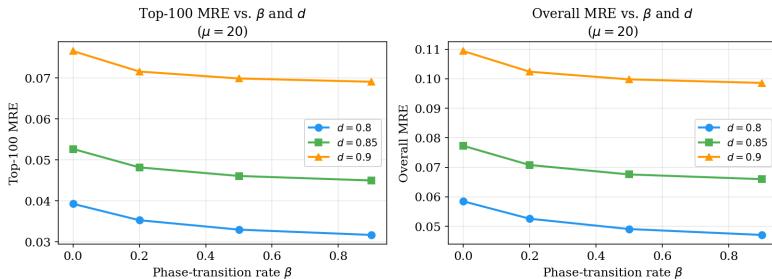
# MRE Heatmap: $\mu$ vs $\beta$



Best combination:  $\mu = 20, \beta = 0.90, d = 0.80$

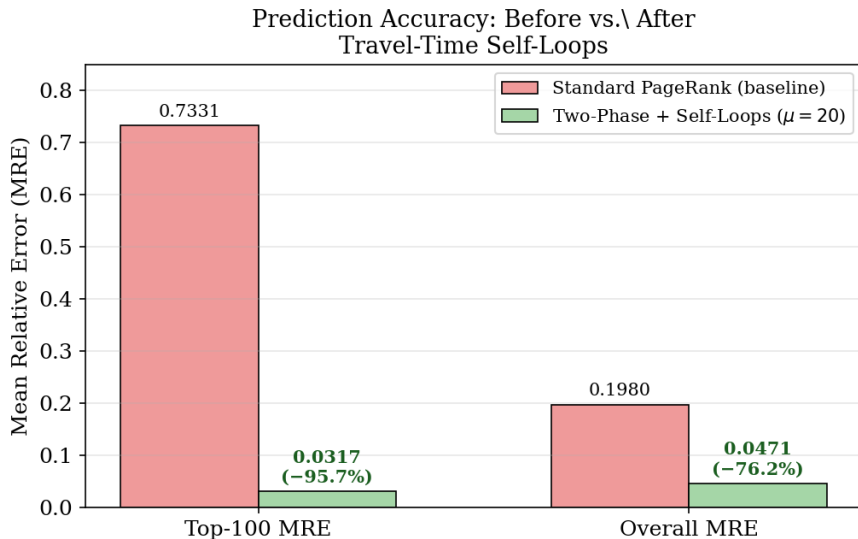
# Damping Sensitivity

Sensitivity Analysis: Damping Factor  $d$  and Phase Rate  $\beta$



$d = 0.80$  balances topology-driven flow with ground-truth injection

# Before & After Self-Loops



# The Four Tuneable Parameters

| Symbol   | Name       | Value | Controls                            |
|----------|------------|-------|-------------------------------------|
| $\gamma$ | Smoothing  | 0.26  | Evidence propagation strength       |
| $\mu$    | Dwell-time | 20    | Self-loop weight (road physics)     |
| $\beta$  | Phase-drop | 0.90  | Active $\rightarrow$ dispersal rate |
| $d$      | Damping    | 0.80  | Topology vs. ground truth           |

## Transferability

For a new city: get road network + some GPS data  $\rightarrow$  grid-search over  $(\gamma, \mu, \beta, d)$

# Key Takeaways

- 1 **Transition matrix**  $M_{2N}$  uses **only topology**  
(road graph + lengths + speeds  $\rightarrow$  two-phase block structure)
- 2 **Teleportation vector**  $E_{2N}$  uses **only ground truth**  
(GPS counts  $\rightarrow$  Laplace  $\rightarrow$  smoothing  $\rightarrow$  normalise)
- 3 **Smoothing justified** by cross-validation:  
 $\gamma^* = 0.26$  minimises MSE (interior minimum, U-shaped curve)
- 4 **Model achieves** MRE = **0.19**  
(37,800 $\times$  improvement from smoothing alone)

# Thank You

Questions?